Surface fault slip from a video camera: a test on the 2025 M_w 7.7 Myanmar earthquake

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Summary. Using CCTV video of the Sagaing fault rupturing in front of the camera, we measured the slip across the fault frame by frame (30 frames/s). The slip in pixels matches the published analyses to within a few percent: one slip pulse of ∼1.3 s and ∼2.5 m. Our open-source GPU code (`slipcam`) finds the slip automatically: on a laptop GPU, decoding the 27-s clip takes 51 s and the automatic pipeline 24 s. A camera measures something seismic instruments do not: the time history of the slip *across* the fault at the surface, with both sides in one image, and it does not saturate at meters of slip. What it lacks (an absolute scale, accurate timing, a stiff mount) can be fixed for a camera installed for this purpose, at ∼$500–1,500 per site.

The test

The video was recorded by a fixed CCTV camera at a solar farm near Thazi (1280×720 pixels, 30 frames/s; a public re-upload). Each frame was correlated with a frame from before the shaking, in 2,450 windows of 64×64 pixels: 667 frames on a laptop RTX 4060 GPU. The camera shakes and rolls, so the slip is measured as the motion of a target on the far side of the fault relative to a camera-side target next to it in the image; their difference removes the camera's own motion. An automatic version (onset, text mask, sides and targets) gives the same pulse with no hand-picked regions.

Table 1: Processing time for the 27-s clip (807 frames) on a laptop RTX 4060 GPU.

Step	Time
Decoding the video into frames (`slipcam-decode`)	51 s
Automatic pipeline, frames to slip (`slipcam-auto`)	24 s
of which the correlation of 560 frames, 2,450 windows each	11 s

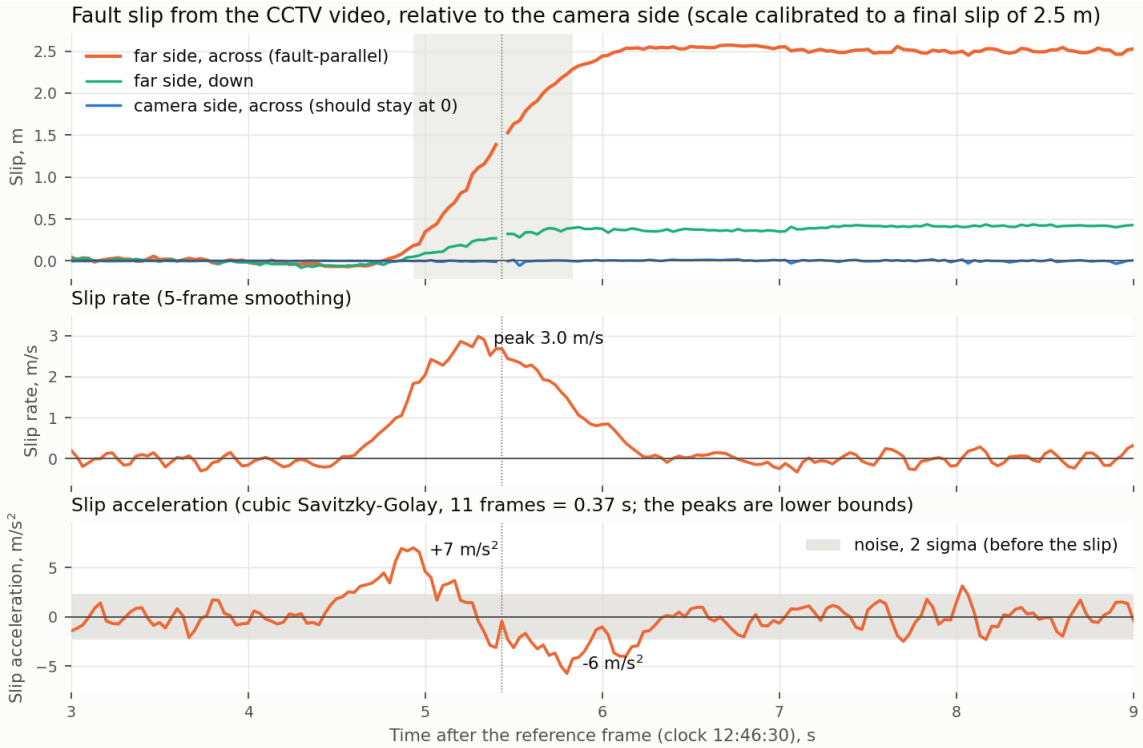


Figure 1: Slip of the far side relative to the camera side (top; the camera side, blue, stays at zero), its rate (middle) and its acceleration (bottom). The slip accelerates at up to ∼7 m/s^2 (∼6σ above the noise) over ∼0.5 s, then decelerates more gradually; smoothing over 0.37 s makes these peaks lower bounds. The scale is set by a final slip of 2.5 m (Kearse & Kaneko, 2025).

The measurement in pixels is consistent across independent codes (Table 2). The spread in meters comes entirely from the scale: each study converted pixels to meters with a different object (a curb, plant pots, a fence, satellite imagery). The noise is ∼0.25–0.3 px per frame, ∼2 cm at this camera's ∼7 cm/pixel.

Table 2: Our measurement compared with the four published analyses of the same video.

	This test	Published
Final offset	35.5 px	35.5 px (Hirano, 2026)
Peak slip rate	39–42 px/s	39.4 px/s (Hirano)
Duration of the slip	∼1.2–1.3 s; one pulse, peak in its first half	1.3–1.4 s, pulse-like (Hirano: ∼2 s)
Peak rate / final slip	1.10–1.19 s^{-1}	1.11–1.28 s^{-1} (all four studies)
Small reverse motion before the slip	∼0.75 px (∼5 cm), 2.5× the frame noise	also reported by Gao et al. and Latour et al.
Final slip, peak rate in meters	2.5 m (set), 2.8–3.0 m/s	1.9–3.0 m, 2.1–3.6 m/s, depending on the scale reference

What it can do now, and what it cannot

- **Can:** recover the slip history at a fault crossing (final offset, slip rate and acceleration, rise time, pulse shape, small motions before the slip) from existing video, with nothing installed beforehand. The automatic pipeline also times the weak first motion (∼6 s before the slip), the stronger shaking and the slip.
- **Needs care:** each target needs its own scale (surveyed points, objects of known size, satellite imagery); one camera cannot separate horizontal from vertical motion in the image plane; consumer clocks drift (this one was ∼4 min off); motion blur and dust lower the quality during and after the slip.
- **Not yet:** video that starts during the shaking, broadcasts, or phones filming a screen. Without retuning, the pipeline partly worked for Hualien 2018 and failed for Kumamoto 2026 (it needs better stabilization).

How a camera compares with conventional instruments

	Camera across the fault	**Strong-motion accelerometer**	**High-rate GNSS**	**Creepmeter**	**InSAR / optical satellite**
Measures	motion of one side relative to the other, in the image plane	ground acceleration at one point	displacement of one point, 3 components	displacement across the fault, along one line	static displacement map
Slip time history at the fault	**yes, directly**	no: one side; static offset lost to integration drift and tilt	one side; stations are rarely within a few km	yes, if the slip stays within its range	no (static only)
Sampling	30–120 frames/s	100–200 samples/s	1–10 samples/s	usually minutes	days to weeks
Resolution	∼2 cm here; mm with a closer view	excellent	∼1 cm	sub-mm	mm–cm (InSAR); dm (optical)
Meters of slip	no saturation (set by the search range)	on scale	on scale	likely off scale or broken	InSAR loses coherence near the rupture
Calibration and timing	**weak**: needs surveyed targets and a GPS-timed clock	calibrated, GPS time	calibrated, GPS time	calibrated	calibrated
Latency	minutes, with streaming or processing on site	seconds	seconds	minutes	hours to days
Rough cost per site	$0.5–1.5k; existing public cameras free	∼$10k+	∼$10k+	a few $k+, with a trench	mission-funded
Limits	light, weather, a stiff mount	—	sky view	needs a known fault trace	revisit time, weather (optical)

The camera adds to seismic instruments; it does not replace them. It fills the gap at the fault itself. An accelerometer records one side and loses the static offset when integrated; GNSS stations are seldom near the fault, and InSAR loses coherence close to the rupture; a creepmeter is built for mm–cm of creep. A camera sees both sides at 30+ frames/s and does not saturate at meters of slip.

The camera is *not* a seismometer: its own shaking is dominated by its mount (a 4.5–5 Hz resonance here), includes rotation, and is not calibrated. Side by side, a GPS-timed camera at a strong-motion station would tie the slip history to the shaking, and one at a creepmeter would calibrate each against the other.

What would make it an instrument

- **Scale and timing:** surveyed targets on both sides of the fault, in view (the largest uncertainty in all published analyses), and a GPS- or NTP-disciplined clock with time stamps for each frame.
- **Camera:** a stiff mount, a global shutter (fast motion skews rolling-shutter images), infrared for night, and a local buffer that survives the loss of power or network.
- **Sites:** lifeline crossings of the San Andreas and other major faults, ideally at USGS creepmeter or strong-motion sites, so every record is paired with a conventional one.

Published analyses of the same video: Kearse & Kaneko (2025), *The Seismic Record* 5(3), 281–288; Latour et al. (2025); Gao et al. (2025); Hirano (2026). **Code:** `slipcam` (MIT license), with a C++/CUDA correlator.